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Deep Kronecker hashing generative adversarial network for intrusion detection in IoT against adversarial attacks

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Abstract

The rapid growth of wireless networks accelerates the adoption of the Internet of Things (IoT) devices in applications, such as industrial automation, healthcare, and smart homes, ensuring seamless connectivity and real-world data transmission. However, the widespread adoption of interconnected IoT systems makes them main targets for cyber threats, particularly adversarial attacks, which exploit system vulnerabilities by manipulating data and compromising privacy and security. Therefore, intrusion detection systems (IDSs) play a vital role in monitoring IoT network behavior and maintaining data leakage, malicious attacks, and unauthorized access. Owing to the heterogeneous nature and resource-constrained characteristics of IoT devices, intrusion detection against adversarial attacks a challenging task. To tackle these challenges, this article develops a novel Deep Kronecker Hashing Generative Adversarial Network (DKHashGAN)-based intrusion detection against adversarial attacks in IoT scenarios. Initially, a Kitsune-based Network Intrusion Detection System (NIDS) is simulated to generate network traffic data. The input log file is normalized by Comprehensive normalization, followed by feature selection based on the Chord distance. Finally, intrusion detection is conducted using the developed DKHashGAN model, which combines a Deep Kronecker Network (DKN) with a Hashing Generative Adversarial Network (HashGAN) to improve detection capability. The robustness of DKHashGAN is evaluated against Projected Gradient Descent (PGD), Basic Iterative Method (BIM), and Fast Gradient Sign Method (FGSM) attacks using K-fold and the percentage of perturbation. Experimental outcomes indicate that the developed model achieves the optimal accuracy, True Positive Rate (TPR), True Negative Rate (TNR), and False Positive Rate (FPR) of 96.43%, 95.90%, 97.91%, and 2.08%, for the FGSM attack using K-fold of 9.

Clinical trial number

Not applicable.

Keywords Internet of things, Comprehensive normalization, Chord distance, Deep Kronecker network, Hashing generative adversarial network

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1 Introduction

In real-world environments, IoT facilitates data communication by connecting physical devices through the internet. Integrated sensors enable these devices to collect and transfer data, enhancing functionality across sectors such as healthcare, agriculture, education, commerce, and defense. It includes a significant contribution to various applications in society, such as medicine, farming, and surveillance. IoT is specified as an interconnected system for transferring data. Hence, IoT will develop over the next few years and solve the network security threats [1–3]. The increasing deployment of IoT technologies in smart-city applications further increases the volume and complexity of network traffic, making effective intrusion detection essential for maintaining secure and reliable services. In large-scale and heterogeneous IoT environments, IDS face challenges in achieving high detection accuracy while minimizing false alarms [4]. There will be an advanced reward for effective IoT breaches, which offers attackers to use more inventive ways for exploiting IoT systems [5]. Still, the traditional approaches and tactics for defending against cyberattacks on the internet are insufficient to protect against the unique security vulnerabilities found in the IoT [6, 7]. As a result, security researchers and experts develop more effective security solutions for solving the security risks in IoT [8, 9]. The multilayered architecture of IoT also exposes different communication and service layers to potential attacks, while resource-constrained devices make the deployment of computationally intensive security mechanisms challenging. Traditional IDS approaches may also exhibit limitations in detecting previously unseen attacks in such dynamic IoT environments [10]. Controlling security risks in every network necessitates three major strategies, such as prevention, identification, and mitigation. All three strategies are needed in order to obtain an effective solution for IoT security [11, 12]. Furthermore, intrusion detection enhances IoT security by monitoring the network layer, identifying suspicious activities, and issuing alerts during cyberattacks [13, 14].

The intrusion detection for an IoT system must have the capability of validating the data packets and generating responses in real time. Moreover, it validates the data packets using different protocols and uses diverse technologies. The intrusion detection is intended for IoT-based smart applications and must operate under complex scenarios, such as quick response, high-volume data processing, and inadequate processing capability. Hence, intrusion detection is combined with protection systems, encryption, and authentication to improve resilience against cyberattacks in IoT networks.

It distinguishes the benign and malicious operations via the utilization of a variety of attack-specific criteria and benign traffic patterns. Furthermore, the intrusion detection system is employed for identifying network threats and preserving cybersecurity [15–17]. In addition to conventional network attacks, the growing deployment of connected devices also facilitates the propagation of IoT botnets, in which compromised devices are coordinated to perform malicious activities and generate abnormal network behavior. The heterogeneous nature and behavioral complexity of IoT botnets make their identification a significant challenge for IoT security [18]. Still, adversarial attack intentionally varies data in order to manipulate the intrusion detection system and avoid security measures. In IoT, a massive volume of networked devices creates an enormous volume of data. Hence, attackers exploit weaknesses in the intrusion detection model by feeding synthetically generated data. These network attacks have occurred in numerous forms, such as the alteration of sensor readings or the introduction of malicious traffic,

which led to misclassification. Most of the IoT systems depend on rule-based intrusion detection or Machine Learning (ML) [19]; hence, adversarial attacks attempt to exploit faults in these models and create false negative results. The varied or dynamic nature of IoT networks is protective against such attacks and ensures the reliability of IoT [20].

Recently, Artificial Intelligence (AI) has allowed computers and other devices for learning from a dataset with minimal human involvement. The subfields of AI are ML and deep learning (DL) methods, where ML offers a satisfactory outcome with less training data, and it distinguishes the attack variants and threats. Moreover, ML-based detection methods are based on domain knowledge, making them simple to design. DL is a subset of ML that provides improved performance over traditional ML methods. It automatically learns features from raw data and effectively processes large volumes of data to generate accurate detection results. One unique feature of DL [21] is its deep structure, which is made up of numerous hidden layers. Traditional ML models, like Support Vector Machine (SVM) are termed as shallow modes, which do not contain any hidden layers. The Neural Network (NN) of the DL model can extract the relevant features from a dataset and then carry out the intrusion detection [8]. IoT security is considered as a significant and continual issue, which requires an up-to-date understanding of security vulnerabilities in IoT, and the development of related mitigation approaches [22]. Traditional ML-based intrusion detection solutions are insufficient for IoT security owing to their resource restrictions and complexity. Hence, the DL model is employed as an effective solution for IoT security, which protects the IoT model from cyber security attacks, and intrusion [23].

1.1 Problem statement and motivation

Existing ML and DL-based IDSs offer potential results based on IoT environments; however, they remain vulnerable to sophisticated adversarial attacks. Conventional feature extraction and detection techniques often fail to capture complex attack patterns and are susceptible to manipulated data, minimizing detection reliability. Moreover, the heterogeneous nature, resource constraints, and dynamic communication behavior of IoT devices limit the scalability and real-world applicability. Hence, an effective intrusion detection mechanism should enable robust feature representation and accurate identification of malicious activities within IoT networks. To address these limitations, this work introduces a novel DKHashGAN model to effectively detect intrusions in IoT against adversarial attacks. Initially, the Kitsune-based NIDS model is simulated, followed by normalization of the input log file using the Comprehensive Normalization technique. The relevant features are then selected based on Chord Distance, after which DKHashGAN is employed to detect intrusions in IoT networks. \DKN and HashGAN are integrated to form the proposed DKHashGAN model.

The major contribution of the work is shown below:

- **DKHashGAN-based intrusion detection in IoT against adversarial attacks:** A novel hybrid detection approach named DKHashGAN is introduced by integrating DKN and HashGAN to improve nonlinear feature learning and enhance robustness against adversarial attacks in IoT networks.
- **Comprehensive normalization for data normalization:** A comprehensive normalization technique is employed to normalize network traffic features based on

their statistical characteristics, reduce scale differences, and mitigate the influence of outliers, thereby improving the quality of the input data.

- **Chord distance for feature selection:** An efficient feature selection mechanism based on Chord Distance is utilized to identify discriminative network traffic features, thereby reducing feature dimensionality and computational complexity while improving intrusion detection performance.
- **Kitsune-based NIDS simulation for realistic traffic generation:** A Kitsune-based NIDS is simulated to capture and extract representative IoT network traffic features under normal and attack scenarios. The extracted traffic features are subsequently processed and provided to DKHashGAN for intrusion detection and classification.

The rest of the paper is organized as follows. Section 2 reviews the existing approaches, while Sect. 3 presents the IoT system model and the Sect. 4 presents the proposed DKHashGAN-based intrusion detection mechanism. Section 5 discusses the experimental results, and Sect. 6 concludes the paper.

2 Literature review

B. Susilo, et al. [24] developed the Autoencoders-Long Short-Term Memory - Convolutional Neural Network (AE-LSTM-CNN) for detecting intrusion in IoT. This model identified numerous security threats and minimized the difficulty of detecting cyberattacks. This model did not explore the robustness and adaptability of intrusion detection across IoT environments with varied data types. S. Balaji and S. Sankara Narayanan [25], devised the Dynamic Distributed Generative Adversarial Network with Improved Firefly Optimization- Hybrid Deep Learning based Convolutional Neural Network-Adaptive Neuro-Fuzzy Inference System (DD-GAN with IFFO-HDLCNN+ANFIS) for intrusion detection in IoT. In this model, the computational complexity was lower. However, the model was not employed as a lightweight distributed approach for applying the end devices of IoT. K. Kostas, et al. [26] developed the IoTGeM-based intrusion detection in IoT. In this model, feature selection was done using the external feedback of an independent dataset, and the data leakage was eliminated. However, the effectiveness of the model remains unvalidated on datasets characterized by complex and intricate data patterns. A. Awajan [27], developed the Deep Learning-based Intrusion Detection (DID) using Deep Neural Network (DNN). This model was more effective and automatically detected intrusion. Nonetheless, this model was not applicable to massive datasets with different IoT devices.

M. Huang, et al. [28] developed the Adversarially Robust IoT Early Defense (ARIoT-EDef) system. This model was effective to detect black-box infection attacks and offered better robustness against adversarial attacks. However, this approach failed to consider host information, like system calls, resources, and memory usage to improve the performance. A. Alotaibi and M. A. R. [29], developed the adversarial perturbation elimination with generative adversarial (APE_GAN++) for intrusion detection in IoT. The model achieved high resilience and sustainability, but variations in training epochs and activation functions were not considered for reducing system complexity. X. Yuana, et al. [30] devised the Deep Learning-based Intrusion Detection System (DLL-IDS). This framework was applicable in various scenarios and achieved significant improvement in performance. Nevertheless, this model failed to focus on the security issues in other domains. S. Hore, et al. [31] developed the One-shot transfer learning approach for

detecting the intrusion in IoT. This model offered improved adaptability to novel attacks while preserving its proficiency in identifying known attack patterns. Still, this model was not validated by human security analysts for detecting and labeling novel attacks

Bajpai, S., et al. [32] designed the Whale optimization method (WOM) to identify the idel parameters for an itrusion detection in IoT and the parameters selected by WOM were employed by the Extreme Gradient Boosting (XGBoost) to enhance intrusion detection accuracy. However, the optimization strategy resulted in higher computational requirements and longer training time. Singh, A.P., et al. [33] introduced a Stacking-enabled Ensemble Learning-based Intrusion Detection Scheme (SELIDS) for Internet of Vehicles (IoV). The tehnique significantly detected multiple cyberattacks based on stacking ensemble of ML algorithms and achieved detection accuracy, but its performance was analyzed a benchmark datasets, limiting its evalution in real-time IoV platforms. Bajpai, S., et al. [34] proposed a cross-silo Federated Learning (FL)-based intrusion detection approach in IoT networks that esured collaborative model training while maintaining data privacy. The approach employed the IoTID20 dataset and obtained high intrusion detection within local and LAN environments. However, the technique did not consider heterogeneous IoT data distributions the scalability and robustness of the technique agsnist cyber threats.

2.1 Major challenges

The major challenges associated with existing IoT intrusion detection methods are summarized as follows:

- The DD-GAN with IFFO-HDLCNN+ANFIS model in [25] offered superior outcomes for intrusion detection. However, it failed to preserve the original relations between features while normalizing data.
- In [27], the DNN was used to provide maximum service quality for intrusion detection. However, it failed to estimate the angular similarity between feature vectors to distinguish the irrelevant and relevant features.
- The model developed in [28] detected the persistent attacks against IoT and enhanced the robustness of the detector. Nevertheless, it failed to increase the detection rates in minority-class attacks.
- In [29], APE_GAN++ was used to attain high accuracy against all types of these attacks. Nevertheless, the processing of the whole dataset required more time and demanded more resources.
- The IoT interrelates individuals with every real-life application, which creates a smart environment via the connectivity of the internet. Still, the IoT devices are affected by security attacks. To overcome these security challenges, several methods have been implemented to detect network intrusion. Owing to the heterogeneous nature and resource constraints of IoT devices, intrusion detection against adversarial attacks becomes complex. To address these issues, a hybrid DL-based intrusion detection is developed.

3 System model for adversarial attack on IoT

The system model [35] for an adversarial attack on IoT is portrayed in Fig. 1. In the model, the Kitsune with a learning-based NIDS identifies the anomalies via the network traffic. The Kitsune comprises five components, such as packet capturing, packet

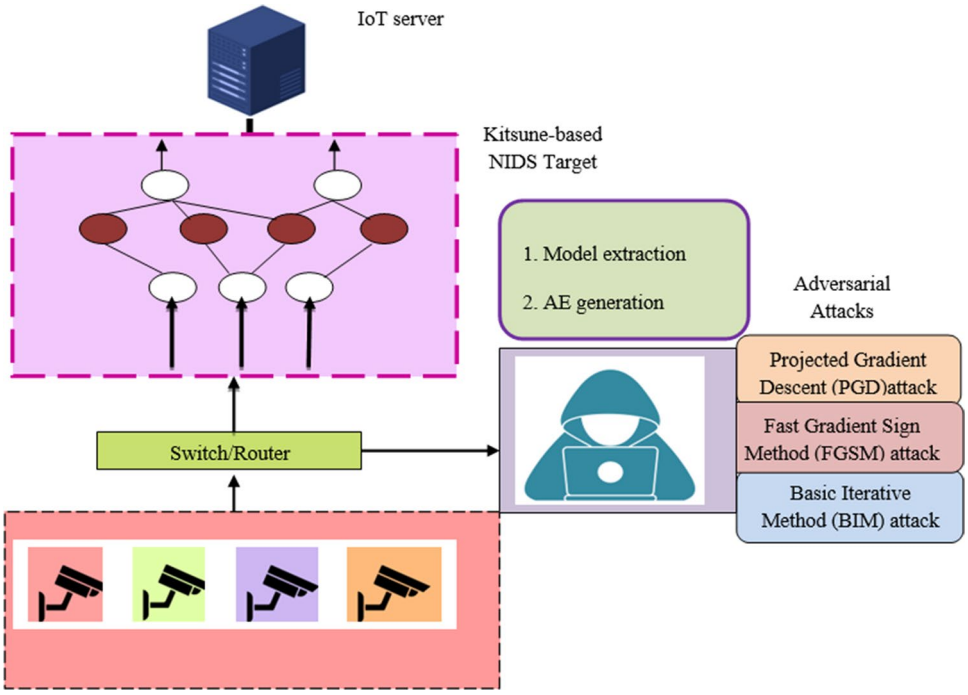


Fig. 1 System model for an adversarial attack on IoT

parsing, feature extraction, Feature Mapping, and Anomaly Detection. Furthermore, Kitsune utilizes unsupervised learning for identifying intrusions on IoT. Initially, the attacker must recreate both the feature extraction and anomaly detection phases using a surrogate shadow model, assuming sufficient knowledge of the feature-processing and anomaly-detection mechanisms of the target NIDS. Afterward, the Adversarial Examples (AEs) are created from the extracted shadow model using the Projected Gradient Descent (PGD), Basic Iterative Method (BIM), and Fast Gradient Sign Method (FGSM) attacks. These attacks apply deliberate perturbations to specific network traffic features while maintaining the essential properties of the original traffic. The packets generate the saliency map, which is considered as an adversary. The saliency map highlights influential features in the detection decision and facilitates their selection for targeted perturbation. Following that, it detects the ideal features with the smallest adjustments and modifies the detected outcome with better success rates while retaining the effects of the original traffic flow on the target system. The attacker aims to perturb the selected features in a manner that misleads the target NIDS into making an erroneous detection decision, resulting in malicious traffic being classified as benign. The attacker utilizes a gradient-based approach for creating the AEs over the specified features and exploits them to compromise the target. Thus, the considered threat model represents an adversarial evasion setting in which the attacker generates carefully perturbed network traffic through the surrogate model and evaluates its ability to evade the target intrusion detection system.

4 Proposed deep Kronecker hashing generative adversarial network-based intrusion detection in IoT against adversarial attack

IoT devices have become an essential component in numerous applications in various sectors. Still, cyberattacks cause malicious activities, impact communication among interconnected devices, and compromise data secrecy. However, intrusion detection in IoT faces some complexities, mainly from adversarial attacks. The adversarial attacks change the functionality of intrusion detection models and cause the network to yield inaccurate outcomes or unnoticed intrusions. The adversarial assaults are resource-constrained, with limited memory, storage, and computational power. Attackers can manipulate these defects by generating hostile inputs and creating wrong identification or misclassification. To overcome these challenges, DKHashGAN is developed as an advanced framework for intrusion detection in IoT networks. Initially, the Kitsune-based NIDS is simulated, and then the input log file taken from the dataset [36] is normalized using the Comprehensive normalization [37]. In this proposed model, Kitsune is primarily used for packet capturing, packet parsing, and network traffic feature extraction, whereas its original anomaly detection ensemble is not used for the final intrusion decision. The extracted feature groups are subsequently normalized and subjected to Chord-distance-based feature selection before being provided to DKHashGAN. Thus, DKHashGAN replaces the anomaly detection stage of Kitsune and performs the final intrusion detection and classification. Thereafter, feature selection is carried out by the chord distance [38], and intrusion detection is performed by the DKHashGan. Figure 2 presents the schematic architecture of the DKHashGAN-based intrusion detection mechanism designed for IoT networks.

4.1 Data acquisition

The input data are collected from the Mirai Raw Dataset [36], which includes 35 raw network traffic samples in the form of PCAP files. The dataset comprises three specific attack samples and five normal traffic samples for cybersecurity analysis and research.

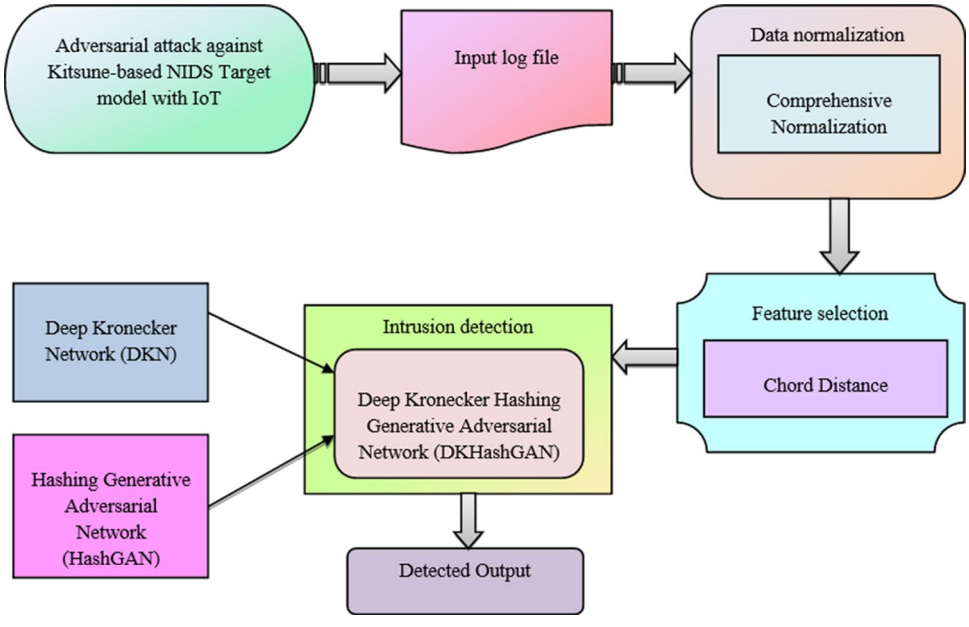


Fig. 2 Schematic depiction of DKHashGan-based intrusion detection in IoT

These files contain an in-depth exploration of IoT botnets via the real-world Mirai attack patterns. Moreover, it yields a valuable ground truth to develop and estimate network-based intrusion detection. Here, the dataset is given by,

$$B = \{B_1, B_2, \ldots, B_c, \ldots, B_u\} \tag{1}$$

The whole data is portrayed by u and c^{th} data is specified by B_c.

4.2 Data normalization using comprehensive normalization

Normalization is a preprocessing technique used to improve data integrity and minimize redundancy. It is also employed as a scaling function to convert the data to a specific standard, eliminate the unit differences, and transform the values to a certain range. Here, the input data B_c is normalized using Comprehensive Normalization [37], which converts all features from the data to a standard value using the statistical characteristics. Moreover, comprehensive normalization is used for eliminating scale differences among the features and minimizing the outlier's impacts. Using Comprehensive Normalization, the normalized data D_c is obtained as follows,

$$D_c = 1 - Exp \frac{\left|(B_c)_{\alpha\beta} - \mathfrak{R}_\beta\right|}{-\left(Max\left\{\underset{\alpha}{Max}\ (B_c)_{\alpha\beta} - \mathfrak{R}_\beta\right\} - Min\left\{\underset{\alpha}{Min}\ (B_c)_{\alpha\beta} - \mathfrak{R}_\beta\right\}\right)} \tag{2}$$

where, $(B_c)_{\alpha\beta}$ shows the input data with an alternative α in criteria β, and the reference value identified by the criteria β is specified by $\mathfrak{R}_\beta$. Furthermore, the normalized data is specified by D_c with the dimension $[\varepsilon \times \eta]$.

4.3 Feature selection by Chord distance

Feature selection is performed to eliminate the distracting parts from the data and minimize the dimensionality of the data. Here, the similarity of the target and the detected output is estimated using the Chord distance [38]. Moreover, the Chord Distance is used as a non-parametric factor for computing the direct relation between the data, and the features are selected from the normalized data D_c with the dimension $[\varepsilon \times \eta]$. Moreover, the Chord distance δ_{Chard} is expressed as follows,

$$\delta_{Chord} = \sqrt{2 - 2.\frac{\sum \varsigma_c \sigma_c}{\sqrt{\sum {\varsigma_c}^2} + {\sigma_c}^2}} \tag{3}$$

Here, the Chord Distance is denoted by δ_{Chord}, and the candidate and target features are indicated by ς_c, and σ_c.

The Chord Distance between each feature and the target is first calculated, after which features exhibiting lower distance values are selected as relevant features. Here, the feature selected is denoted by L_c with the size $[\varepsilon \times \gamma]$, where $\eta > \gamma$. Table 1 shows the selected features in the feature selection process.

4.4 Intrusion detection in IoT using proposed DKHashGAN

IoT environments are susceptible to diverse cyber threats, such as data breaches, malware attacks, and unauthorized access. Intrusion detection plays a vital role in protecting the integrity and confidentiality of IoT systems against such threats.Hence, the

Table 1 Selected features

SL. No	Features
1	ether_src
2	ether_dst
3	ether_type
4	ip_id
5	ip_flag
6	ip_ttl
7	tcp_dport
8	tcp_window
9	tcp_chksum
10	ip_vers
11	ip_ihl
12	ip_len
13	ip_prot
14	ip_chks
15	ip_src
16	ip_dst
17	tcp_sport
18	tcp_se
19	tcp_ack
20	tcp_dataofs

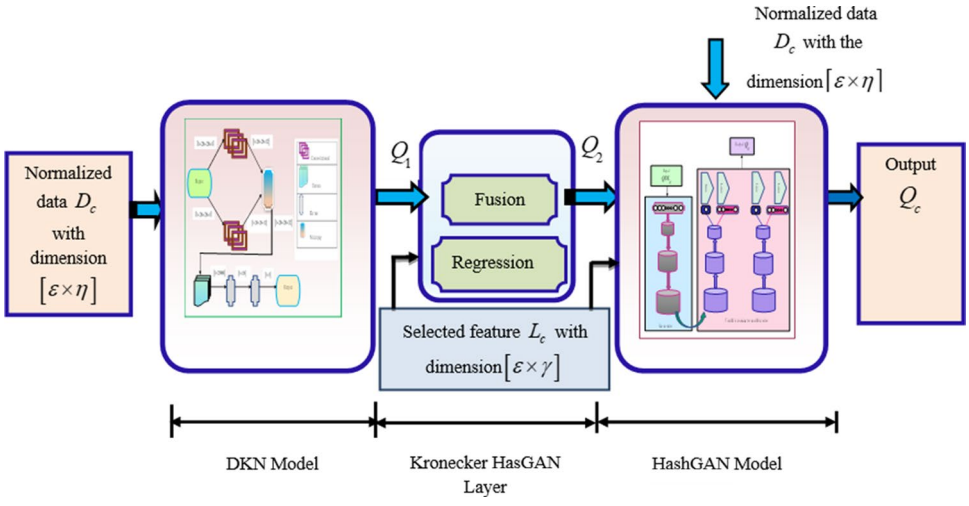


Fig. 3 Structure of DKHashGAN

DKHashGAN is dev eloped for intrusion detection. The merging of DKN [39] with HashGAN [40] develops the DKHashGAN. Fig. 3 shows the schematic view of DKHashGAN, where the normalized data D_c with the dimension $[\varepsilon \times \eta]$ is fed to DKN to obtain the outcome Q_1. The output of DKN (Q_1) and the selected feature L_c with dimension $[\varepsilon \times \gamma]$ are given to the Kronecker HasGAN layer, where fusion and regression create the outcomeQ_2. Finally, the outcome Q_2, normalized data D_c of size $[\varepsilon \times \eta]$ and feature L_c with dimension $[\varepsilon \times \gamma]$ are merged as a single term $QDL_c \in \{Q_2, D_c, L_c\}$ and then QDL_c is applied to the HashGAN to obtain the detected outcome Q_c. Moreover, the Adam Optimizer trains the hyperparameters of DKHashGAN.

4.4.1 DKN

The Kronecker product is considered as the major parameter of DKN [39], such as iwhere the layersnput, hidden, and output as well as the block matrix are presented. The structure of DKN is displayed in Fig. 4, where normalized data D_c of size $[\varepsilon \times \eta]$ is fed as the input, and the outcome of DKN is denoted by Q_1.

$$Q_1 = \xi\left(o_j\right) Exp\left\{o_j\left\langle\left(D_c\right)_j, S\right\rangle - \gamma\left(\left\langle\left(D_c\right)_j, S\right\rangle\right)\right\} \tag{4}$$

The term Q_1 specifies the outcome of DKN, and the known univariate functions are indicated by $\xi(.)$ and $\gamma(.)$. Moreover, certain known links function $f(.)$ is given by,

$$f\left(X\left(o_j\right)\right) = \left\langle\left(D_c\right)_j, S\right\rangle \tag{5}$$

Using the coefficient S and rank U, the Kronecker product decomposition becomes $T(\geq 2)$.

$$S = \sum_{u=1}^{U} W_T^u \otimes W_{T-1}^u \otimes \ldots \otimes W_1^u \tag{6}$$

Furthermore $t = 1, \ldots, T$, and $u = 1, \ldots, U$ are represented as the Kronecker component, where $m = \Pi_{t=1}^{T} m_t$ and $\xi = \Pi_{t=1}^{T} \xi_y$ is obtained as follows.

$$W_{t''} \otimes W_{t''-1} \otimes \cdots \otimes W_{t'''} = \bigotimes_{t=t'''}^{t''} W_t \tag{7}$$

For every matrix, the decomposition is portrayed by $S = \sum_{u=1}^{U} \otimes_{t=T}^{1} W_t^u$.

4.4.2 Kronecker HasGAN layer

In Kronecker HasGAN layer, output of DKN (Q_1) and the feature L_c with size $[\varepsilon \times \gamma]$ are considered as the input. Here, the outcome Q_2 is generated using the fusion and regression process. The outcome of Kronecker HasGAN layer at the interval r is given by,

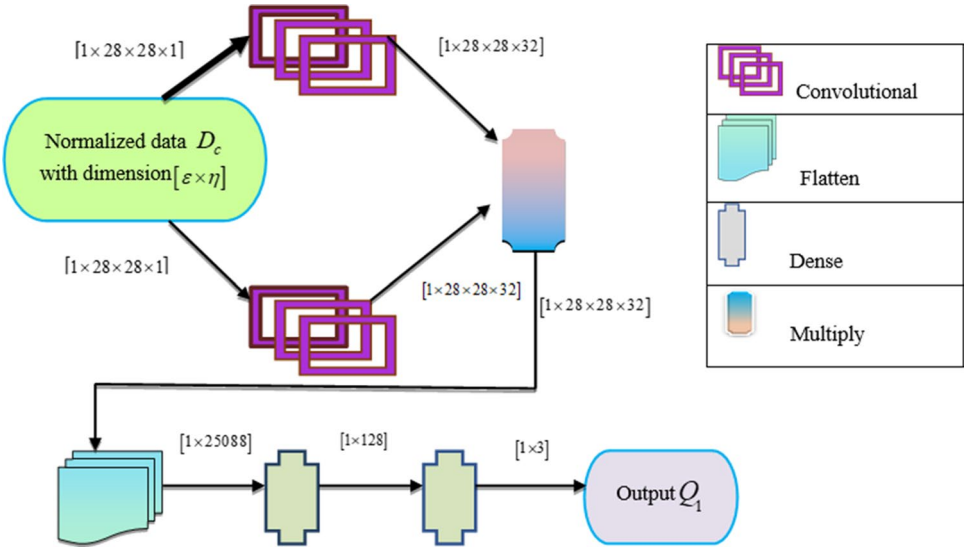


Fig. 4 Schematic depiction of DKN

$$G(r) = \sum_{e=1}^{E} (L_c)_e * w_e \tag{8}$$

The term $G(r)$ denotes the outcome at the interval r, and the selected features are specified by L_c. The weight is denoted by w and E implies the entire count of features.

Using the Fractional Calculus (FC) [41], the output of Kronecker HasGAN layer (Q_2) at $(r+1)$ interval is given by,

$$G(r+1) = q.G(r) + \frac{1}{2}q.G(r-1) \tag{9}$$

Consider $G(r+1) = Q_2$ and $G(r-1) = Q_1$. Hence, Eq. (10) becomes,

$$Q_2 = q.\sum_{e=1}^{E} (L_c)_e * w_e + \frac{1}{2}q.\left[\xi(o_j)\, Exp\left\{o_j\left\langle (D_c)_j, S\right\rangle - \gamma\left(\left\langle (D_c)_j, S\right\rangle\right)\right\}\right] \tag{10}$$

The output of the Kronecker HasGAN layer is denoted by Q_2 and q specifies the constant.

4.4.3 HashGAN

HashGAN [40] comprises three parts such as generator, encoder, and discriminator. Although HashGAN is originally developed for unsupervised image retrieval and hashing, the proposed DKHashGAN adapts its hashing capability to network intrusion detection by using the generated binary codes as compact discriminative representations rather than directly treating them as class labels. The outcome of the Kronecker HashGAN Layer (Q_2), along with the selected future L_c with the dimension $[\varepsilon \times \gamma]$ and the normalized data D_c with the dimension $[\varepsilon \times \eta]$ are combined as a single term $QDL_c \in \{Q_2, D_c, L_c\}$, which is applied as the input of HashGAN. The encoder is used for mapping the input to discriminative binary hash codes. In the proposed DKHashGAN, these binary hash codes are treated as compact discriminative representations of the learned network traffic features rather than as direct intrusion class labels. The pictorial view of HashGAN is shown in Fig. 5, where the concatenation of samples from the uniform and binary random variables is fed to the generator. For training the parameters in the discriminator, the adversarial loss function is utilized. Moreover, the encoding parameters are trained using hashing loss, and the generator's parameters are trained by a feature mapping loss. In the HashGAN, the encoder and discriminator are tied to obtain the deep hash function with unsupervised training. The encoder pathway is used for data distribution that is determined in the hidden variables of the discriminator. The interpolated input space of generator creates semantic dissimilarities in data distribution; therefore, the encoder learns the biased binary codes via the training. The feature matching loss along with ℓ_2 norm loss trains the encoder and generator. The HashGAN uses a multilayer hash encoder for mapping the input data into hash codes with A bits. Therefore, A independent sigmoid activations are presented in the final layer of tencoder, and the output of the encoder is specified by $b_d = \mu(QDL_c)$. The resulting binary hash representation is subsequently forwarded to the classification component of DKHashGAN, where the learned hash representation is mapped to the corresponding intrusion category through the class-probability outputs. The class corresponding to the maximum predicted probability is assigned as the final class label, enabling the discrimination of normal traffic from malicious and adversarial traffic classes. Hence, the encoder-generated binary hash codes function as

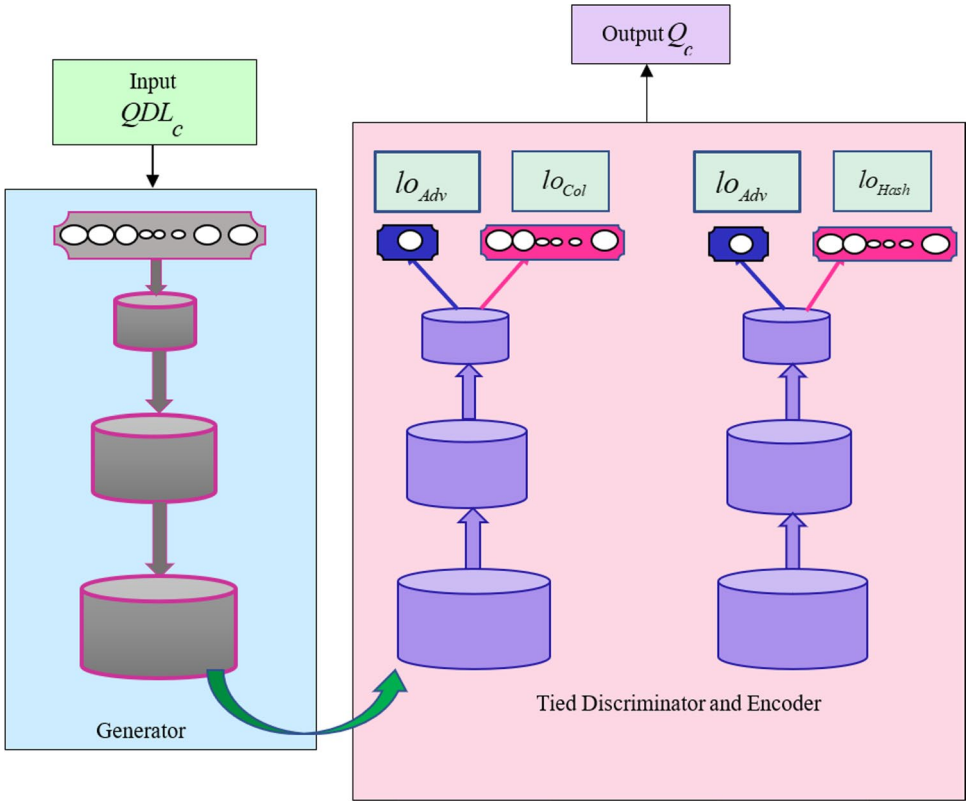


Fig. 5 Structure of HashGAN

discriminative feature representations, while the subsequent classification operation performs the actual intrusion-label assignment. Furthermore, A independent probability is denoted by,

$$b_{Da} = \Pr\left(\lambda_{Da} = 1 | QDL_c; \varpi_\mu\right) \tag{11}$$

where, b_{Da}, and λ_{Da} shows the ath outcome of the encoder as well as the binary code of dth data. The encoding parameter is denoted by ϖ_μ, the binary hash codes are estimated by setting $\lambda_{Da} = 1|\left(b_{Da} > 0.5\right)$, and $1\left(.\right)$ represents the indicator function. The generator network in the HashGAN is used for mapping the samples to the data distribution, and a random input of the generator is concatenated with the uniform and binary random variables as $h_d = \left[h'_d, \lambda'_d\right]$, where the binary random noise is specified by $h'_d \sim F$, and the uniform random noise is denoted by $\lambda'_d \sim I$. The final layer of the discriminator is represented as a sigmoid activation, where the probabilities are given by,

$$\rho_d = M\left(\left(QDL_c\right)_d\right) = \Pr\left(z_d = 1 | \left(QDL_c\right)_d; \varpi_M\right) \tag{12}$$

$$\hat{\rho}_d = M\left(J\left(QDL_c\right)_d\right) = \Pr\left(z_d = 1 | \left(\hat{QDL}_c\right)_d; \varpi_M\right) \tag{13}$$

The probability of real ($z_d = 1$) and synthesized data are specified by ρ_d, and $\hat{\rho}_d$. The final loss function (Lo_{tl}) of the HashGAN is obtained as the summation of hashing (lo_{Hash}), collaborative (lo_{Col}), and adversarial (lo_{Adv}) losses.

$$lo_{tl} = lo_{Adv} + lo_{Col} + lo_{Hash} \tag{14}$$

The detected output (Q_c) of HashGAN is obtained by,

$$Q_c = fn(Q_2) \tag{15}$$

$$Q_c = fn\left(q.\sum_{e=1}^{E}(L_c)_e * w_e + \frac{1}{2}q.\left[\xi(o_j)\,Exp\left\{o_j\left\langle(D_c)_j, S\right\rangle - \gamma\left(\left\langle(D_c)_j, S\right\rangle\right)\right\}\right]\right) \tag{16}$$

where, Q_2 specifies the output of the Kronecker HashGAN Layer.

The Pseudocode for DKHashGAN is given in Algorithm 1.

Algorithm 1. Pseudocode for DKHashGAN

DKHashGAN
Input
Training: Normalized data D_c^{Train} , Feature L_c^{Train}, and training labels
Testing: Normalized data D_c^{Test} , Feature L_c^{Test} , and training labels
Output
Bias and weight of DKHashGAN
Output of DKHashGAN (Q_c)
Parameters Used
Highest Epoch $\left(Mx_{epo}\right)$
Learning rate
(lr) Batch Size (bs)
Start
Initialization
Weight and bias of DKHashGAN
Initialize bs, lr, and Mx_{epo}
//**Training**
while $Epoch < Mx_{epo}$
For every training data
Execute DKN, the output Q_1 is, $Q_1 = \xi(o_j)\,Exp\left\{o_j\left\langle(D_c)_j, S\right\rangle - \Upsilon\left(\left\langle(D_c)_j, S\right\rangle\right)\right\}$
Using fusion and regression, the outcome of Kronecker HashGAN layer $\left(Q_2\right)$ $Q_2 = q.\sum_{e=1}^{E}\left(L_c\right)_e * w_e + \frac{1}{2}q.\left[\xi(o_j)\,Exp\left\{o_j\left\langle(D_c)_j, S\right\rangle - \Upsilon\left(\left\langle(D_c)_j, S\right\rangle\right)\right\}\right]$
Using HashGAN for yielding the final output Q_c $Q_c = fn\left(q.\sum_{e=1}^{E}\left(L_c\right)_e * w_e + \frac{1}{2}q.\left[\xi(o_j)\,Exp\left\{o_j\left\langle(D_c)_j, S\right\rangle - \Upsilon\left(\left\langle(D_c)_j, S\right\rangle\right)\right\}\right]\right)$
End for
4 Calculate errors Err using Q_c, and training data labels
5 Update bias and weight by Adam optimizer.
$Epoch$
End while
Testing
For each test data D_c^{Test}
Compute Q_1 *and* Q_2 by the trained Kronecker HashGAN layer
Estimate the label Q_c
End for
End

5 Results and discussion

This section discusses the experimental evaluation of DKHashGAN-based intrusion detection in IoT networks against adversarial attacks. The datasets, implementation environment, evaluation metrics, and comparative performance analysis are also presented.

5.1 Experimental setup

The DKHashGAN-based intrusion detection framework is implemented in a Python environment to evaluate its performance against adversarial attacks in IoT networks. The experimental parameters used for the evaluation are listed in Table 2.

5.2 Dataset description

The data are collected from the Mirai Raw Dataset [36], which comprises 35 network traffic samples representing normal and Mirai-based DDoS attack scenarios. These traffic samples are provided through 89 raw PCAP files containing the corresponding network traffic captures. The dataset includes five normal-traffic samples and three samples for each specific Mirai DDoS attack type, thereby providing diverse traffic patterns for IoT cybersecurity analysis. The raw PCAP files contain real-world network traffic generated during normal and Mirai attack scenarios and provide valuable ground-truth information for developing and evaluating network-based IDS.

Table 2 Parameters

Parameter	Value
Attack Type	PGD, BIM, FGSM
Threat Model	White-box
Perturbation Norm	L_∞
Perturbation Budget (ε)	0.08 (8%)
Step Size (α)	0.008
Attack Iterations	20
Maximum Optimization Steps	100
Random Initialization	Uniform (−0.08, 0.08)
Random Restarts	10
Batch Size	64
Learning Rate	0.001
Gradient Computation	Backpropagation
Gradient Update Frequency	Every Iteration
Gradient Clipping Threshold	1.0
Projection Method	Projection onto L_∞-ball
Input Clipping Range	[0, 1]
Confidence Threshold	0.95
Decision Threshold	0.50
Early Stopping Tolerance	$\left(\frac{1}{times\ 10^{\{-5\}}}\right)$
Momentum Factor	0.90
Weight Decay	$\left(\frac{1}{times\ 10^{\{-4\}}}\right)$
Floating-Point Precision	FP32
Random Seed	42
Runtime Budget	120 s/sample
Confidence Interval	95%

5.3 Performance computing metrics

Metrics such as accuracy, TPR, TNR, and FPR are employed to evaluate the effectiveness of DKHashGAN.

5.3.1 Accuracy

Accuracy [42] is specified as the fraction of exact predictions (both normal and intrusions) to the overall predictions.

$$Accuracy = \frac{H_{PO} + K_{PO}}{H_{PO} + K_{PO} + H_{NE} + K_{NE}} \tag{17}$$

Here, H_{PO} and K_{PO} indicates the true and false positives. Furthermore, H_{NE} and K_{NE} symbolizes the true and false negatives.

5.3.2 TPR

TPR [42] is specified as the count of actual intrusions that are correctly identified as positive. Furthermore, the TPR is expressed by,

$$TPR = \frac{H_{PO}}{H_{PO} + K_{NE}} \tag{18}$$

5.3.3 TNR

TNR [42] is represented as the proportion of normal activities, which are identified as normal. The following expression denotes the Specificity.

$$TNR = \frac{H_{NE}}{H_{NE} + K_{PO}} \tag{19}$$

5.3.4 FPR

FPR measures the proportion of normal activities incorrectly classified as intrusions. A lower FPR indicates fewer false alarms and better detection reliability. The FPR is expressed as follows,

$$FPR = \frac{K_{PO}}{K_{PO} + H_{NE}} \tag{20}$$

5.4 Simulation results

Figure 6 depicts the simulation output of DKHashGAN-based Intrusion detection in IoT. Figure 6 (a) illustrates the deviation plot for values of −1, 0, and 1, where 0 represents accurate detection and deviations within the range of − 1 to 1 indicate inaccurate detection.

Figure 6 b) depicts the prediction checkpoint graph for intrusion detection. Here, the densely distributed samples in class 1 accurately detect the intrusion, whereas the dispersed samples indicate that the detection is incorrect. The denser samples in Class 0 wrongly detect the intrusion, whereas the dispersed samples accurately detect the intrusion.

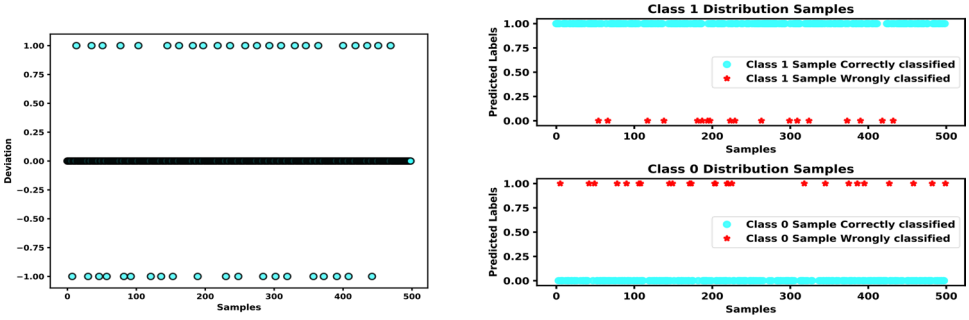


Fig. 6 Experimental result for DKHashGAN-based intrusion detection in IoT, **a**) deviation graph, **b**) prediction checkpoint graph

5.5 Comparative methods

The methods like DD-GAN with IFFO-HDLCNN+ANFIS [25], DNN [27], ARIoT-EDef [28], APGAN++ [29], Transformer-based IDS [43], Dynamic Defense Selection Framework (DYNAMITE) [44], and Adversarially Optimized Multi-Space Prototypical Networks (MSPL-IDS) [45] are considered as the existing methods for validating the effectiveness of DKHashGAN-based Intrusion detection in IoT.

5.6 Comparative assessment

The performance of DKHashGAN-based intrusion detection in IoT is comparatively evaluated under different adversarial attacks, namely PGD, BIM, and FGSM.

5.6.1 Comparative assessment using PGD attack

DKHashGAN-based intrusion detection under the PGD attack is evaluated using different K-fold values and perturbation percentages.

a) Assessment based on K-Fold

Figure 7 depicts the performance of DKHashGAN for different K-fold values. The assessment in terms of accuracy is shown in Fig. 7 a). Consider the K-Fold = 5, the accuracy of 90.75% is attained by the DKHashGAN, and the values 82.84%, 83.34%, 83.56%, 85.55%, 85.83%, 86.66%, and 87.51% are achieved by the existing models. Accordingly, DKHashGAN demonstrates performance gains of 8.715%, 8.165%, 7.920%, 5.418%, 5.730%, 4.507%, and 3.568% over the considered existing methods. Fig. 7 b) displays the assessment concerning TPR. The DKHashGAN attains a TPR is 90.01%, and the values 81.82%, 82.11%, 82.77%, 84.25%, 84.86%, 85.72%, and 82.29% are achieved by the traditional approaches. Accordingly, DKHashGAN demonstrates performance gains of 9.098%, 8.777%, 8.045%, 6.399%, 5.722%, 4.766%, and 3.015% over the considered existing models. The estimation in terms of TNR is depicted in Figure 7 c). Considering the K-Fold of 5, the TNR of 82.30%, 83.21%, 85.33%, 86.51%, 86.81%, 87.08%, 87.35%, and 91.04% is achieved by the classical techniques and the proposed DKHashGAN. Hence, 9.604%, 8.601%, 6.276%, 4.976%, 4.646%, 4.350%, and 4.053% of improved performance are achieved by the DKHashGAN compared to the baseline methods. Figure 7 d) illustrates the FPR performance of DKHashGAN under different K-fold configurations. For a K-fold value of 5, the baseline models achieve FPR values of 17.7%, 16.79%, 14.66%, 13.49%, 13.18%, 12.92%, and 12.64%, whereas the proposed DKHashGAN achieves an FPR of 8.95%.

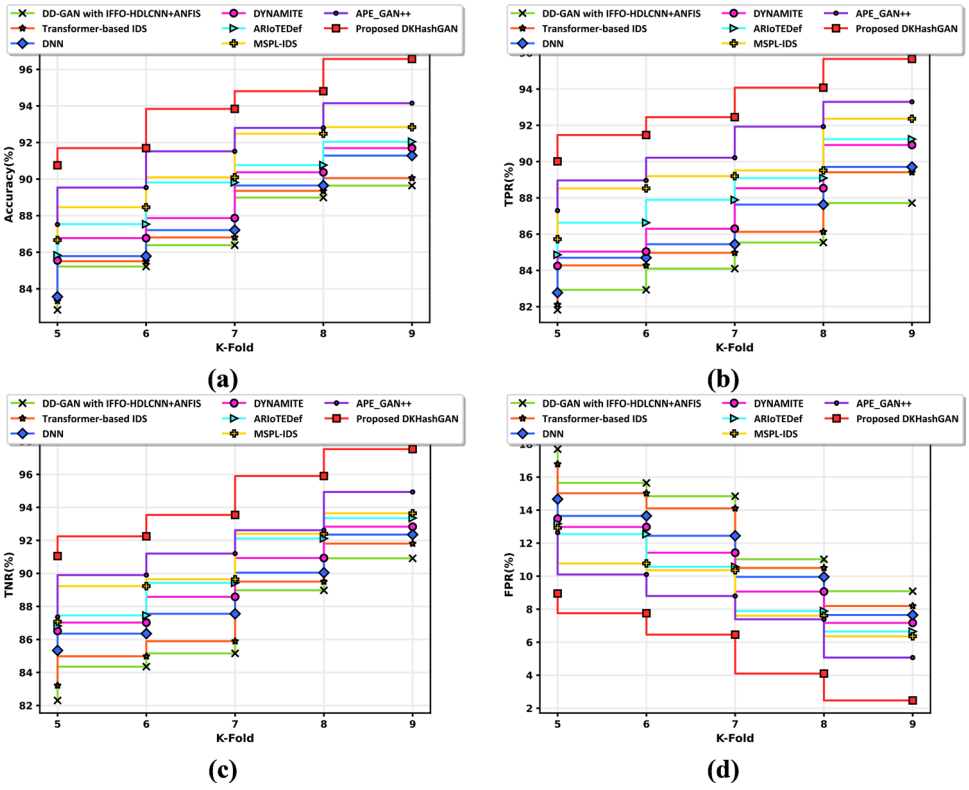


Fig. 7 Comparative assessment of DKHashGAN using K-Fold, **a**) accuracy, **b**) TPR, **c**) TNR, and **d**) FPR

b) Assessment using Percentage of perturbation

Figure 8 shows the estimation of DKHashGAN using Percentage of perturbation. The evaluation concerning accuracy is portrayed in Fig. 8 a). For the Percentage of perturbation = 2%, the DKHashGAN achieved an accuracy of 96.36%, whereas the conventional techniques reach the accuracy of 89.62%, 91.00%, 91.86%, 92.06%, 92.26%, 92.73%, and 94.12%. Hence, 6.995%, 5.562%, 4.670%, 4.462%, 3.767%, and 2.325%, of performance improvement is obtained by the existing models. The assessment in terms of TPR is shown in Fig. 8 b). The TPR of 95.49% is achieved by the DKHashGAN, in which the baseline approaches obtain the TPR of 88.81%, 89.18%, 89.62%, 90.46%, 90.98%, 92.20%, and 92.77%, while considering the Percentage of perturbation of 2%. Hence, 6.995%, 6.608%, 6.147%, 5.268%, 4.723%, 3.445%, and 2.848% of improved performance is attained by the DKHashGAN than the existing methods. The estimation regarding TNR is portrayed in Fig. 8 c). For the Percentage of perturbation of 2%, the TNR of 90.93%, 91.39%, 92.16%, 93.16%, 93.45%, 94.16%, 95.69%, and 97.71% are achieved by the traditional and the developed DKHashGAN models. Therefore, the performance improvement of 6.939%, 6.466%, 5.684%, 4.652%, 4.355%, 3.631%, and 2.068% is recorded by the classical techniques. Figure 8 d) illustrates the evaluation of FPR. At a 2% of the perturbation, the proposed DKHashGAN approach and the baseline techniques reach the FPR of 2.29%, 9.07%, 8.61%, 7.84%, 6.84%, 6.55%, 5.84%, and 4.31%.

5.6.2 Comparative assessment using BIM attack

The comparative evaluation of DKHashGAN-based Intrusion detection in IoT using BIM attack is performed by changing the K-Fold, and Percentage of perturbation.

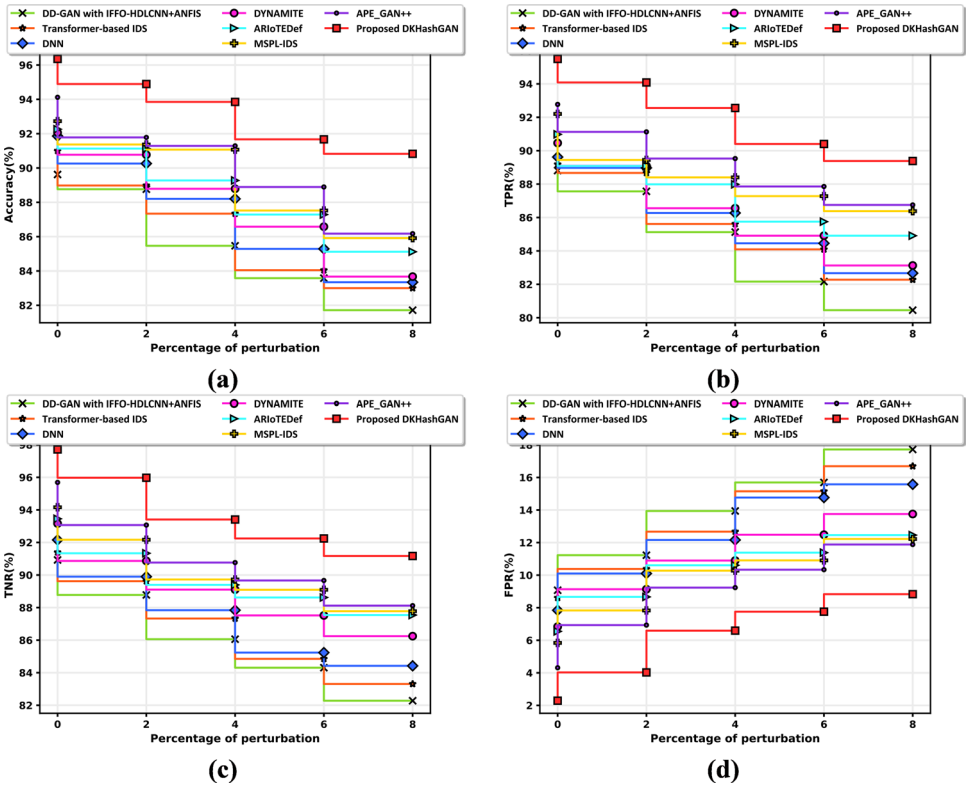


Fig. 8 Comparative evaluation of DKHashGAN using percentage of perturbation, **a**) accuracy, **b**) TPR, **c**) TNR, and **d**) FPR

a) Assessment based on K-Fold

The assessment of DKHashGAN using the variation of K-Fold is shown in Fig. 9. The evaluation regarding accuracy is shown in Fig. 9 a). For the K-Fold = 5, the accuracy of 90.20% is attained by DKHashGAN, whereas the values attained by the traditional approaches is 81.46%, 82.22%, 83.73%, 84.12%, 85.03%, 86.71%, and 87.90%. Therefore, the DKHashGAN has achieved a performance improvement of 9.690%, 8.859%, 7.174%, 6.756%, 5.735%, 3.876%, and 2.565%. The analysis in terms of TPR is portrayed in Fig. 9 b). The TPR of 90.41% is achieved by the DKHashGAN, whereas the existing models achieved the TPR of 81.96%, 82.69%, 83.97%, 84.53%, 85.26%, 86.69%, and 87.66% for the K-Fold of 5. Therefore, the improved performance of 9.346%, 8.543%, 7.123%, 6.500%, 5.694%, 4.118%, and 3.037% is achieved by the classical approaches. The evaluation regarding TNR is portrayed in Fig. 9 c). For the K-Fold of 5, the conventional techniques and the proposed DKHashGAN has achieved the TNR of 82.91%, 83.36%, 83.69%, 84.58%, 87.19%, 87.26%, 87.19%, and 91.21%. Therefore, 9.104%, 8.618%, 8.239%, 7.275%, 4.410%, 4.342%, and 4.403% of improved performance is attained by the DKHashGAN than the existing methods. In Fig. 9 d), the analysis of FPR is demonstrated. At a k-fold of 5, the classical models and the introduced DKHashGAN approach obtain the FPR of 17.09%, 16.64%, 16.30%, 15.42%, 12.81%, 12.74%,12.80%, and 8.78%.

b) Assessment using percentage of perturbation

The analysis of DKHashGAN using the Percentage of perturbation is depicted in Fig. 10. The assessment regarding accuracy is shown in Fig. 10 a). The DKHashGAN has achieved an accuracy of 96.19%, in which the conventional techniques achieve the accuracy of 88.49%, 90.32%, 90.53%, 91.25%, 92.36%, 93.01%, and 94.04%, for the percentage

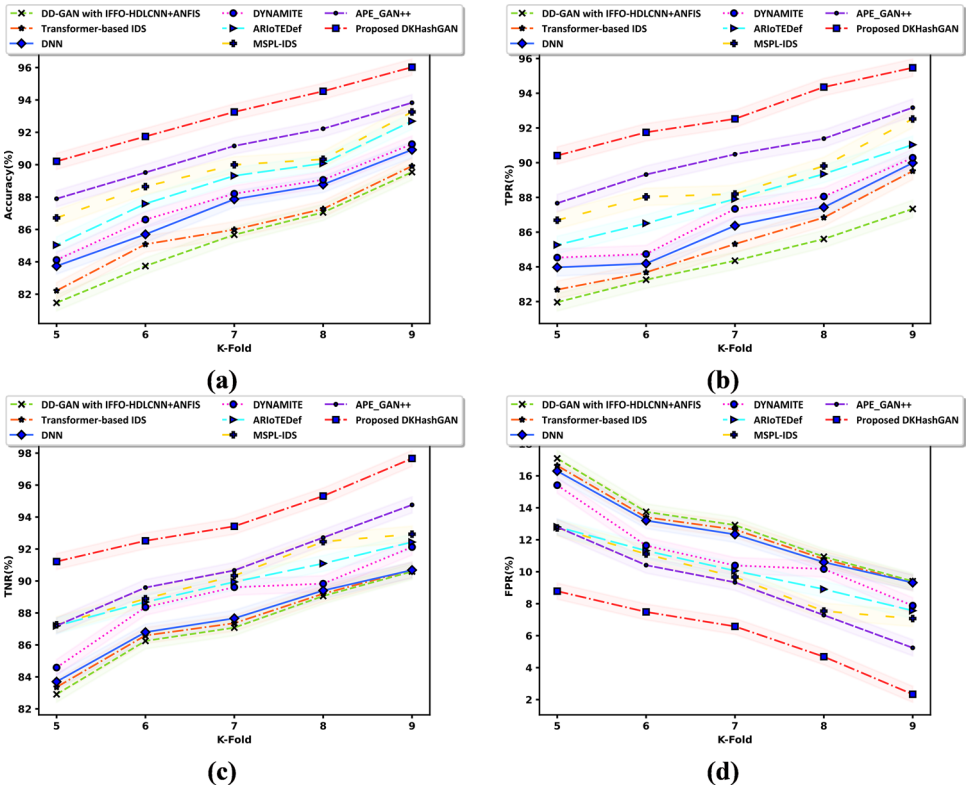


Fig. 9 Comparative assessment of DKHashGAN using K-Fold, **a**) accuracy, **b**) TPR, **c**) TNR, and **d**) FPR

of perturbation of 2%. Hence, 8.005%, 6.098%, 5.880%, 5.139%, 3.977%, 3.308%, and 2.231%, of improved performance is achieved by the traditional methods. Fig. 10 b) deliberates the estimation regarding TPR. While considering the percentage of perturbation of 2%, the TPR of 95.66% is achieved by the DKHashGAN, in which the values recorded by the classical approaches is 87.16%, 88.89%, 89.30%, 90.07%, 92.20%, 92.81%, and 93.58%. Therefore, 8.887%, 7.077%, 6.649%, 5.840%, 3.621%, 2.982%, and 2.177% of performance improvement is attained by the DKHashGAN compared to the existing models. Figure 10 c) portrays the evaluation in terms of TNR. The TNR of 89.54%, 90.41%, 90.69%, 91.94%, 92.16%, 92.37%, 92.89%, and 96.82% is achieved by the baseline techniques and the DKHashGAN while considering the Percentage of perturbation = 2%. Hence, the superior performance of 7.519%, 6.621%, 6.331%, 5.040%, 4.813%, 4.596%, and 4.059%, is attained by the DKHashGAN than the existing methods. In Fig. 10 d), the FPR validation is demonstrated. By considering the percentage of perturbation is 2%, the traditional techniques and the introduced DKHashGAN model achieves the FPR of 10.46%, 9.59%, 9.31%, 8.06%, 7.84%, 7.63%, 7.11%, and 3.18%.

5.6.3 Comparative assessment using the FGSM attack

The comparative evaluation of DKHashGAN-based Intrusion detection in IoT using the FGSM attack is performed by changing the K-Fold and the Percentage of perturbation.

a) Assessment based on K-Fold

Figure 11 deliberates the evaluation of DKHashGAN by varying the K-Fold. The assessment regarding accuracy is portrayed in Fig. 11 a). For the K-Fold = 5, the DKHashGAN has achieved an accuracy of 90.89%, and the values achieved by the

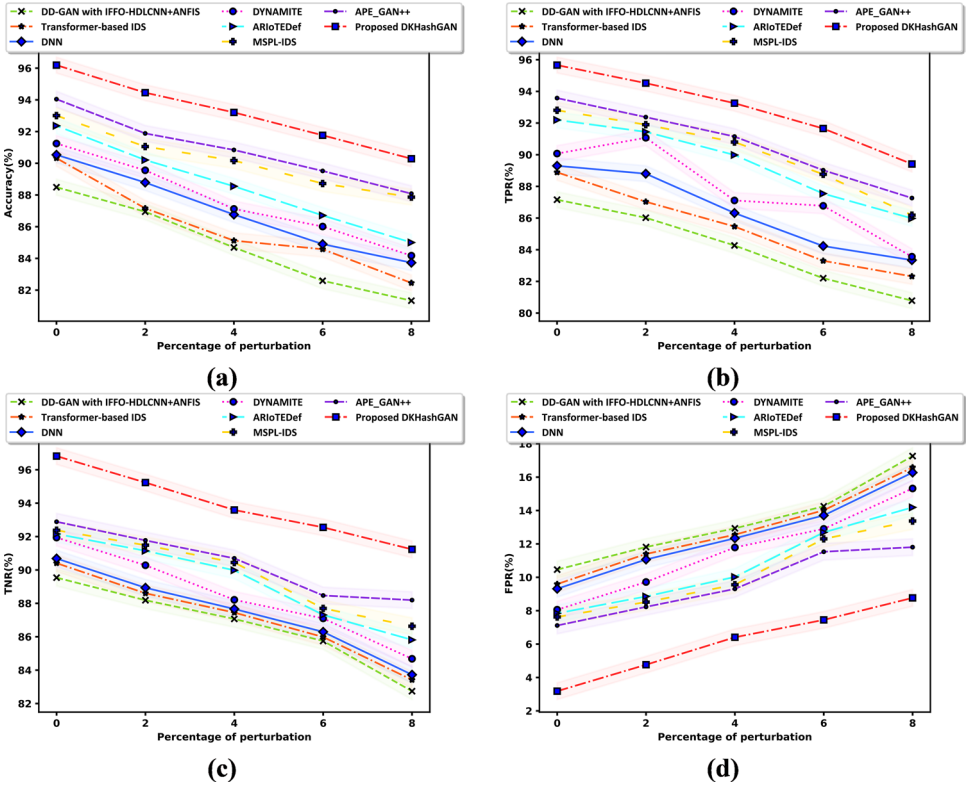


Fig. 10 Comparative evaluation of DKHashGAN using percentage of perturbation, **a**) accuracy, **b**) TPR, **c**) TNR, and **d**) FPR

traditional approaches is 82.28%, 83.15%, 84.59%, 84.95%, 86.18%, 87.83%, and 88.57% for the K-Fold of 5. Therefore, the DKHashGAN has attained an improved performance of 9.473%, 8.526%, 6.931%, 6.546%, 5.191%, 3.377%, and 2.552% than the existing models. Figure 11 b) shows the evaluation regarding TPR. For the K-Fold of 5, the TPR of 89.24% is achieved by the developed DKHashGAN model and conventional techniques reached the values of 81.52%, 82.41%, 82.82%, 84.24%, 84.90%, 85.62%, and 86.32%. Hence, the performance improvement of 8.652%, 7.668%, 7.198%, 5.611%, 4.863%, 4.067%, and 3.268% is attained by the DKHashGAN than the classical approaches. The estimation concerning TNR is portrayed in Fig. 11 c). For the K-Fold of 5, the TNR of 84.00%, 84.89%, 85.17%, 86.41%, 86.80%, 86.71%, 87.88%, and 91.21% are achieved by the baseline techniques and the proposed DKHashGAN. Therefore, the DKHashGAN has obtained an improved performance of 7.900%, 6.931%, 6.620%, 5.262%, 4.828%, 4.933%, and 4.743% than the existing models. Figure 11 d) demonstrates the FPR analysis. When the K-fold of 5, the FPR obtained by the proposed DKHashGAN model is 8.79%; meanwhile, the traditional techniques reach the FPR of 16.00%, 15.11%, 14.83%, 13.59%, 13.19%, 13.29%, and 13.12%.

b) Assessment using the percentage of perturbation

The assessment of DKHashGAN-based Intrusion detection in IoT using the Percentage of perturbation is shown in Fig. 12. The evaluation concerning accuracy is portrayed in Fig. 12 a). For the Percentage of perturbation = 2%, the DKHashGAN has achieved an accuracy of 96.60%, whereas the existing models achieve the accuracy of 89.67%, 90.19%, 90.57%, 92.14%, 92.70%, 93.32%, and 94.51%. Therefore, 7.176%, 6.636%, 6.242%,

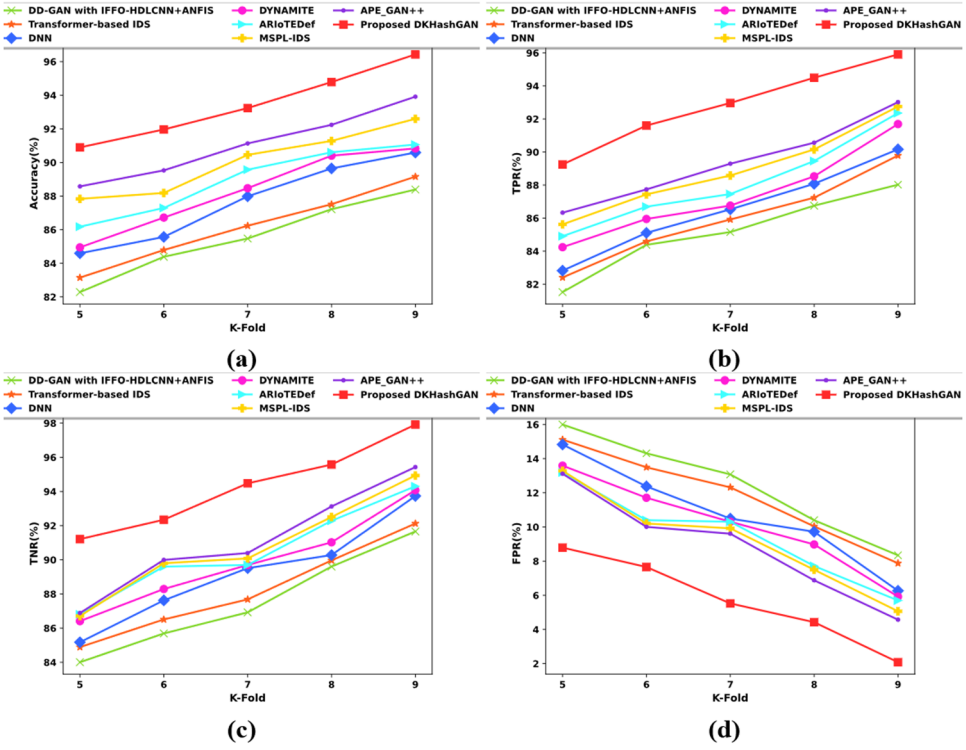


Fig. 11 Comparative assessment of DKHashGAN using K-Fold, **a**) accuracy, **b**) TPR, **c**) TNR, and **d**) FPR

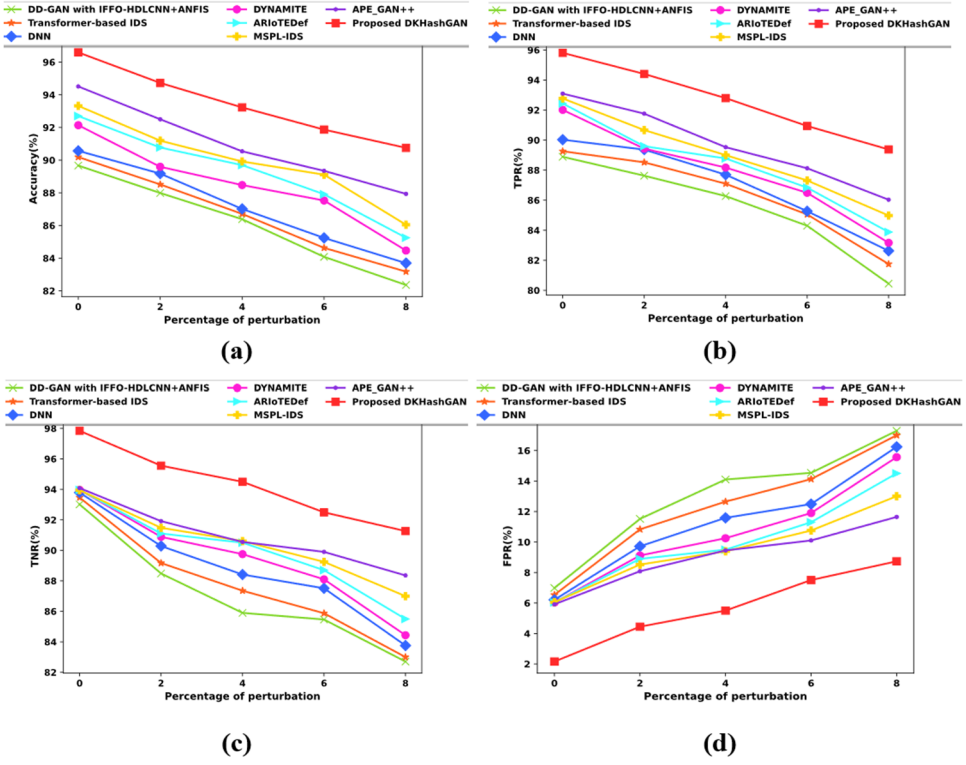


Fig. 12 Comparative evaluation of DKHashGAN using percentage of perturbation, **a**) accuracy, **b**) TPR, **c**) TNR, and **d**) FPR

4.617%, 4.037%, 3.391%, and 2.162%, of better performance is achieved by the DKHashGAN compared to the existing methods. The assessment in terms of TPR is shown in Fig. 12 b). The TPR of 95.81% is achieved by the DKHashGAN, whereas the classical techniques achieve the TPR of 88.89%, 89.26%, 90.03%, 92.01%, 92.44%, 92.77%, and 93.10% while considering the Percentage of perturbation of 20%. Hence, 7.223%, 6.836%, 6.033%, 3.966%, 3.517%, 3.173, and 2.829% of improved performance are achieved by the DKHashGAN compared to the conventional techniques. The evaluation regarding TNR is depicted in Fig. 12 c). Considering the Percentage of perturbation of 20%, the classical models, and the DKHashGAN has obtained the TNR of 93.01%, 93.47%, 93.79%, 93.97%, 93.96%, 93.95%, 94.08%, and 97.84%. Hence, the improved performance of 4.932%, 4.469%, 4.137%, 3.955%, 3.962%, 3.979%, and 3.839% is reached by the DKHashGAN than the existing models. Figure 12 d) presents the FPR analysis. At a percentage of perturbation is 2%, the proposed DKHashGAN model and the traditional techniques reach the FPR of 2.16%, 6.99%, 6.53%, 6.21%, 6.03%, 6.04%, 6.05%, and 5.92%.

5.7 Ablation analysis

Figure 13 presents the ablation analysis for the PGD, BIM, and FGSM attacks by evaluating accuracy across different K-fold settings. The analysis compares the individual components, including DKHashGAN without data normalization, DKHashGAN without feature selection, DKN, and HashGAN, with the complete DKHashGAN model. Figure 13 a) illustrates the ablation results for the PGD attack. When the K-fold is 9, the individual configurations obtain accuracy of 92.790%, 93.498%, 94.526%, and 95.455%, respectively, whereas the proposed DKHashGAN achieves the highest accuracy of 96.332%. Figure 13 b) presents the ablation results for the BIM attack. For the K-fold of 9, the

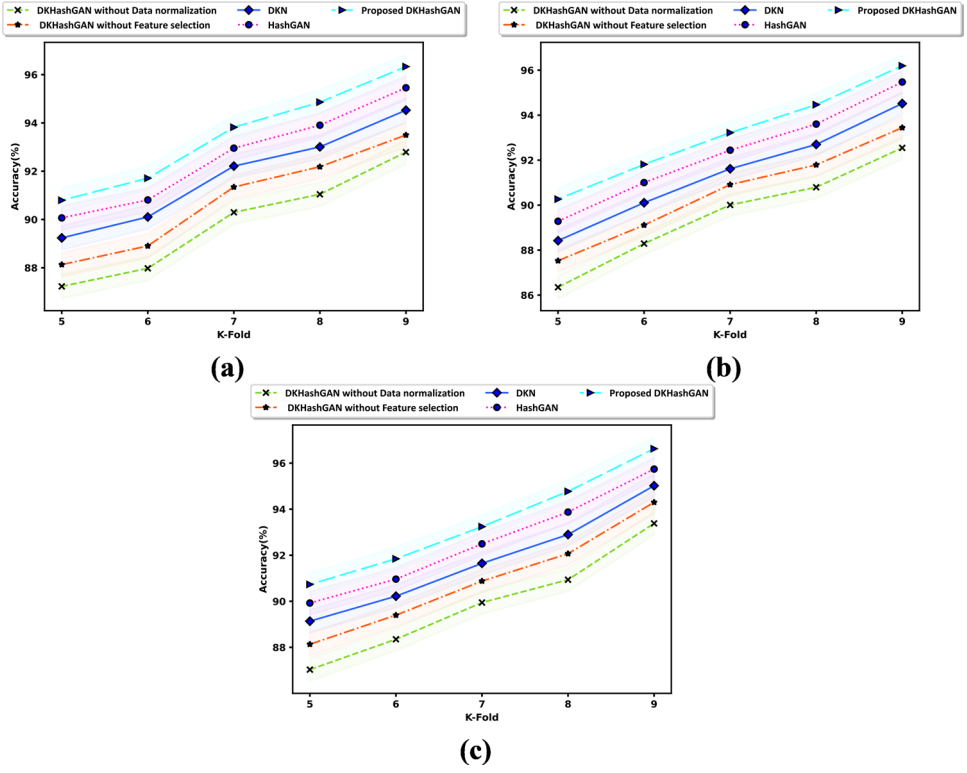


Fig. 13 Ablation analysis for (**a**) PGD attack, (**b**) BIM attack, and (**c**) FGSM attack

respective configurations reach accuracy of 92.545%, 93.444%, 94.516%, and 95.473%, while DKHashGAN achieves the highest accuracy of 96.194%. Figure 13 c) depicts the ablation results for the FGSM attack. At a K-fold of 9, the individual configurations record accuracy of 93.382%, 94.299%, 95.019%, and 95.741%, respectively, whereas the proposed DKHashGAN achieves the highest accuracy of 96.625%. Overall, the ablation results demonstrate that each preprocessing and model component contributes to the overall detection capability of DKHashGAN. The complete integration of data normalization, feature selection, DKN, and HashGAN provides more discriminative feature representations and improves robustness against PGD, BIM, and FGSM attacks compared with the individually configured models.

5.8 Comparative discussion

Table 3 displays the comparative discussion of DKHashGAN, where the assessment is based on K-Folds, and the Percentage of perturbation using three types of attacks like PGD, BIM, and FGSM. The DKHashGAN has attained the optimal values in the FGSM attack using K-Fold variations. The DKHashGAN has achieved the superior accuracy of 96.43%, and the values 88.39%, 89.16%, 90.59%, 90.83%, 91.06%, 92.60%, and 93.91% are obtained by the existing models. The TPR of 95.90% is achieved by the DKHashGAN, in which the traditional approaches obtain a TPR of 88.02%, 89.79%, 90.16%, 91.69%, 92.36%, 92.75%, and 93.02%. The better TNR of 97.92% is achieved by the DKHashGAN, whereas the TNR recorded by the conventional techniques are 91.66%, 92.12%, 93.73%, 94.08%, 94.30%, 94.94%, and 95.43%. The FPR reached by the conventional techniques are 8.34%, 7.88%, 6.27%, 5.92%, 5.70%, 5.06%, and 4.57%, while the proposed DKHashGAN model achieves the lowest FPR of 2.08%. Moreover, the accuracy, TPR, TNR, and FPR of 90.75%, 89.38%, 91.27%, and 8.73% are achieved by DKHashGAN in the Percentage of perturbation -based assessment. Using the PGD attack, the accuracy, TPR, TNR, and FPR of 96.57%, 95.66%, 97.53%, and 2.47%, are reached by the DKHashGAN in K-Fold-based assessment, and the values 90.88%, 89.38%, 91.17%, and 8.83% are obtained by the DKHashGAN in Percentage of perturbation-based assessment. Moreover, the accuracy, TPR, TNR, and FPR of 96.02%, 95.47%, 97.67%, and 2.33%, recorded by the DKHashGAN in K-Fold-based assessment as well as the values 90.29%, 89.41%, 91.23%, and 8.77% are obtained by the DKHashGAN in Percentage of perturbation-based assessment while considering the BIM attack. Here, the proposed DKHashGAN yields the optimal values due to the avoidance of overfitting. The use of K-fold cross-validation reduces the dependence on a single training-testing partition and provides a more reliable assessment of the model's generalization capability, thereby helping to identify potential overfitting. The consistent performance observed across the folds indicates that the proposed model maintains stable detection capability rather than relying on a particular data partition. This robustness is attributed to the complementary capabilities of DKN and HashGAN for adversarial intrusion detection in IoT platforms. DKN accurately models complex relationships between network traffic features through Kronecker-based representation, ensuring the extraction of discriminative and high-dimensional feature structures. However, its robustness against adversarially manipulated samples is limited. HashGAN overcomes these challenges by learning compact hash-based feature embeddings based on adversarial training, which enhances resilience and feature separability to adversarial attacks. The combination of these two mechanisms

Table 3 Comparative discussion of DKHashGAN-based intrusion detection in IoT

Types of Attack	Variations	Metrics/ Methods	DD-GAN with IFFO-HDLCNN+ ANFIS	Transformer -based IDS	DNN	DYNAMITE	ARIoTEDef	MSPL-IDS	APE_ GAN++	Proposed DKHashGAN
PGD attack	*K-Fold = 9*	*Accuracy (%)*	89.64	90.05	91.28	91.69	92.05	92.84	94.15	96.57
		TPR (%)	87.72	89.41	89.71	90.92	91.24	92.36	93.29	95.66
		TNR (%)	90.91	91.81	92.35	92.83	93.35	93.64	94.93	97.53
		FPR (%)	9.09	8.19	7.65	7.17	6.65	6.36	5.07	2.47
	Percentage of perturbation = 90%	*Accuracy (%)*	81.72	83	83.34	83.67	85.12	85.92	86.18	90.83
		TPR (%)	80.45	82.28	82.67	83.13	84.91	86.38	86.75	89.38
		TNR (%)	82.28	83.31	84.42	86.24	87.54	87.78	88.12	91.17
		FPR (%)	17.72	16.69	15.58	13.76	12.46	12.22	11.88	8.83
BIM attack	*K-Fold = 9*	*Accuracy (%)*	89.54	89.91	90.92	91.27	92.69	93.26	93.83	96.02
		TPR (%)	87.34	89.52	89.99	90.28	91.03	92.52	93.17	95.47
		TNR (%)	90.59	90.70	90.68	92.12	92.43	92.93	94.76	97.67
		FPR (%)	9.41	9.30	9.32	7.88	7.57	7.07	5.24	2.33
	Percentage of perturbation = 90%	*Accuracy (%)*	81.33	82.45	83.73	84.17	85	87.87	88.08	90.29
		TPR (%)	80.78	82.32	83.34	83.56	85.98	86.18	87.26	89.41
		TNR (%)	82.74	83.42	83.72	84.68	85.81	86.63	88.19	91.23
		FPR (%)	17.26	16.58	16.28	15.32	14.19	13.37	11.81	8.77
FGSM attack	*K-Fold = 9*	*Accuracy (%)*	88.39	89.16	90.59	90.83	91.06	92.60	93.91	**96.43**
		TPR (%)	88.02	89.79	90.16	91.69	92.36	92.75	93.02	**95.90**
		TNR (%)	91.66	92.12	93.73	94.08	94.30	94.94	95.43	**97.92**
		FPR (%)	8.34	7.88	6.27	5.92	5.70	5.06	4.57	**2.08**
	Percentage of perturbation = 90%	*Accuracy (%)*	82.36	83.18	83.69	84.47	85.25	86.05	87.93	90.75
		TPR (%)	80.45	81.75	82.63	83.16	83.88	84.98	86.03	89.38
		TNR (%)	82.70	83.00	83.76	84.43	85.50	87.00	88.36	91.27
		FPR (%)	17.30	17.00	16.24	15.57	14.50	13.00	11.64	8.73

provides robust feature vector, minimizes feature redundancy, enhances generalization over heterogeneous IoT traffic, and raises detection reliability for evaluating cyber threats. Subsequently, DKHashGAN obtains higher intrusion detection performance while facilitating improved robustness compared with DKN and HashGAN models.

5.9 T-test analysis

A paired t-test is employed to determine whether the performance differences between the proposed method and the baseline approaches are statistically significant based on accuracy, TPR, and TNR. The statistical analysis considers the fold-level outcomes from 9-fold cross-validation (K = 9) for PGD, BIM, and FGSM attacks. Each metric is evaluated by comparing the performance across the nine folds between the proposed method and every baseline model. Statistical significance is established when the p-value is less than 0.05. The obtained results are reported in Table 4. Across the evaluated metrics, the proposed method exhibits statistically significant performance gains over DD-GAN with IFFO-HDLCNN+ANFIS, Transformer-based IDS, DNN, DYNAMITE, ARIoT-EDef, MSPL-IDS, and APE_GAN++. Under the PGD attack condition, all baseline comparisons satisfy the significance criterion of $p < 0.05$. Specifically, the proposed method records higher accuracy than DD-GAN with IFFO-HDLCNN+ANFIS ($t = 2.88$, $p = 0.01$), Transformer-based IDS ($t = 2.61$, $p = 0.02$), DNN ($t = 2.34$, $p = 0.02$), DYNAMITE ($t = 2.04$, $p = 0.03$), ARIoTEDef ($t = 1.77$, $p = 0.04$), MSPL-IDS ($t = 1.45$, $p = 0.04$), and APE_GAN++ ($t = 1.13$, $p = 0.05$). Likewise, significant performance improvements are obtained for TPR

Table 4 T-test analysis

Proposed model	Existing models	T-statistics			p-value		
		Accuracy	TPR	TNR	Accuracy	TPR	TNR
PGD attack							
DKHashGAN	DD-GAN with IFFO-HDLCNN+ANFIS	2.88	2.85	2.84	0.01	0.01	0.01
	Transformer-based IDS	2.61	2.59	2.56	0.02	0.02	0.02
	DNN	2.34	2.32	2.30	0.02	0.02	0.02
	DYNAMITE	2.04	2.03	2.01	0.03	0.03	0.03
	ARIoTEDef	1.77	1.75	1.74	0.04	0.04	0.04
	MSPL-IDS	1.45	1.44	1.42	0.04	0.04	0.04
	APE_GAN++	1.13	1.11	1.10	0.05	0.05	0.05
BIM attack							
DKHashGAN	DD-GAN with IFFO-HDLCNN+ANFIS	2.86	2.84	2.82	0.01	0.01	0.01
	Transformer-based IDS	2.60	2.57	2.55	0.02	0.02	0.02
	DNN	2.32	2.30	2.28	0.02	0.02	0.02
	DYNAMITE	2.03	2.01	2.00	0.03	0.03	0.03
	ARIoTEDe	1.75	1.74	1.72	0.04	0.04	0.04
	MSPL-IDS	1.44	1.43	1.41	0.04	0.04	0.04
	APE_GAN++	1.12	1.10	1.09	0.05	0.05	0.05
FGSM attack							
DKHashGAN	DD-GAN with IFFO-HDLCNN+ANFIS	2.85	2.82	2.80	0.01	0.01	0.01
	Transformer-based IDS	2.58	2.56	2.53	0.02	0.02	0.02
	DNN	2.31	2.29	2.27	0.02	0.02	0.03
	DYNAMITE	2.02	2.00	1.98	0.03	0.03	0.03
	ARIoTEDef	1.74	1.73	1.71	0.04	0.04	0.04
	MSPL-IDS	1.43	1.41	1.40	0.04	0.04	0.04
	APE_GAN++	1.10	1.09	1.08	0.05	0.05	0.05

and TNR, further supporting the effectiveness of the proposed method against PGD attacks. The performance of the proposed method under the BIM attack is further validated through statistical analysis, demonstrating significant improvements over all baseline methods across the evaluated metrics. In comparison with IFFO-HDLCNN+ANFIS, the obtained t-statistics are 2.86, 2.84, and 2.82, respectively, with $p = 0.01$. Significant differences are also observed against the Transformer-based IDS ($p = 0.02$), DNN ($p = 0.02$), DYNAMITE ($p = 0.03$), ARIoTEDef ($p = 0.04$), MSPL-IDS ($p = 0.04$), and APE_GAN++ ($p = 0.05$). Under the FGSM attack scenario, the proposed method exhibits statistically significant improvements over the baseline approaches for all evaluated metrics. Against IFFO-HDLCNN+ANFIS, the corresponding t-statistics reach 2.85, 2.82, and 2.80, respectively, with $p = 0.01$. The statistical significance is likewise maintained in comparisons with the Transformer-based IDS ($p = 0.02$), DNN ($p = 0.02$ and 0.03), DYNAMITE ($p = 0.03$), ARIoTEDef ($p = 0.04$), MSPL-IDS ($p = 0.04$), and APE_GAN++ ($p = 0.05$). The statistically significant and consistent results obtained under PGD, BIM, and FGSM attacks highlight the robustness of the proposed method and demonstrate its ability to achieve improved classification performance over traditional approaches.

6 Conclusion

The increasing adoption of IoT across diverse application domains exposes interconnected systems to security vulnerabilities and adversarial attacks that can manipulate network traffic and compromise the reliability of IDS. To overcome these challenges, the proposed DKHashGAN model is developed for adversarial intrusion detection in IoT environments. Initially, a Kitsune-based NIDS is simulated, after which the input log file undergoes Comprehensive Normalization. Relevant features are subsequently selected using Chord Distance, and intrusion detection is performed using DKHashGAN. The effectiveness of DKHashGAN is evaluated against PGD, BIM, and FGSM adversarial attacks using accuracy, TPR, TNR, and FPR, based on K-fold of 9 and percentage of perturbation. Under the FGSM attack with 9-fold cross-validation, the DKHashGAN achieves an accuracy, TPR, TNR, and FPR metrics and the corresponding values of 96.43%, 95.90%, 97.91%, and 2.08%, demonstrating effective detection performance with a reduced FPR. The consistent performance under different adversarial attack scenarios further indicates the robustness and generalization capability of the proposed model. Overall, DKHashGAN provides an effective approach for strengthening intrusion detection against adversarial attacks in heterogeneous IoT networks. In the future, a hybrid optimization technique will be developed for training the DKHashGAN to achieve even more precise detection.

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Author contribution
SUNITHA A conceived the presented idea and designed the analysis. Also, she carried out the experiment and wrote the manuscript with support from S G HYMLIN ROSE. All authors discussed the results and contributed to the final manuscript. All authors read and approved the final manuscript.

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Data availability
The data that support the findings of this study are openly available in the Mirai raw dataset at "https://www.kaggle.com/datasets/nguyenquanitmo/mirai-raw-dataset".

Code availability
The source code for the proposed method in available at https://github.com/Sunitha2705/DKHashGAN.git

Declarations

Ethical approval
Not applicable.

Informed consent
Not applicable.

Compliance with ethical standards
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